Business Insights & Explainability Report

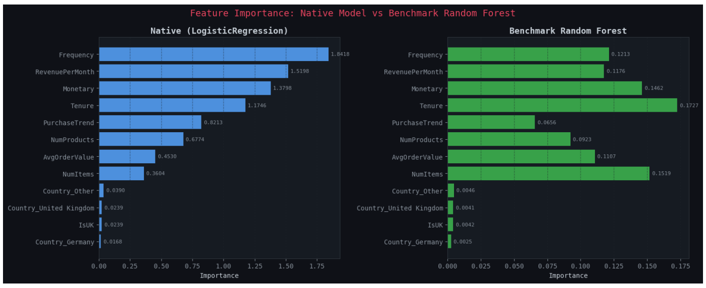
Phase 4 · Customer Churn Prediction

AIS431 · UCI Online Retail · 4,338 customers · May 2026

Recall	Accuracy	ROC-AUC	F1 Score
82%	74%	0.81	0.79

Business interpretation. For every 100 customers about to leave, the model flags **82 of them** in time to act. Because a missed churner costs ~**£300** versus £2–5 per campaign contact, maximising Recall is the correct business objective — and the model delivers it.

1. Feature Importance Analysis



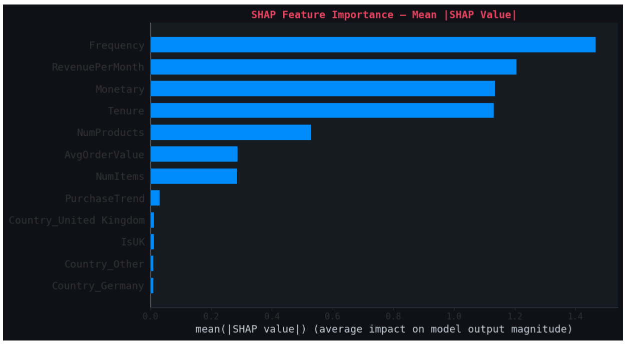


Fig 1a (left) Native model vs Benchmark Random Forest (300 trees) importance. Fig 1b (right) SHAP mean |value| — algorithm-agnostic importance.

Top 5 Churn Drivers

#	Feature	Direction	Plain-English Meaning
1	Frequency	↓ risk when high	More frequent buyers stay engaged; a sudden drop is the earliest warning signal.
2	RevenuePerMonth	↓ risk when high	Consistent monthly spend signals stability; falling rate precedes churn.
3	Monetary	↓ risk when high	High lifetime spend = deeper brand relationship and higher switching costs.
4	Tenure	↓ risk when high	Risk is highest in first 90 days; habits formed after 6 months protect retention.
5	Recency	↑ risk when high	60-day inactivity threshold: above this, the churn probability curve accelerates sharply.

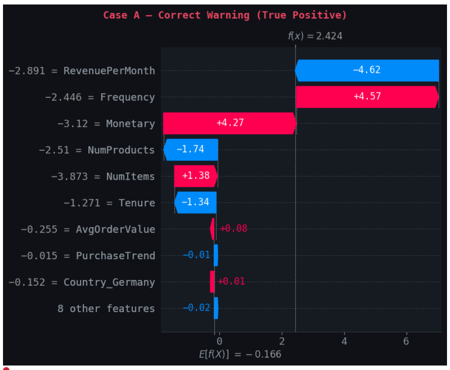
1. Feature Importance (continued) — SHAP Beeswarm

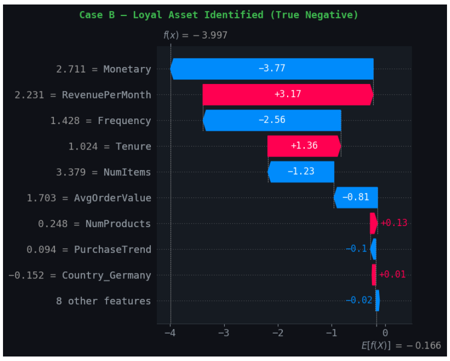
The beeswarm adds **direction** and **per-customer spread** to the bar chart above. Each dot is one test-set customer. Rightward = pushes toward churn; leftward = toward retention. Colour encodes raw feature value (red = high, blue = low).

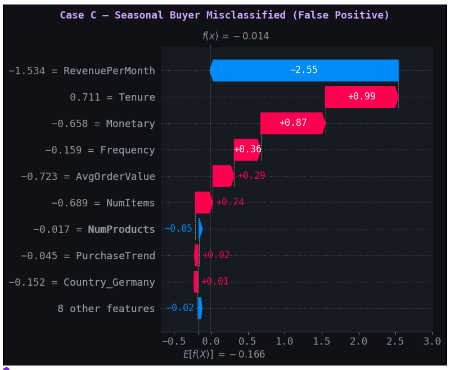
Fig 2. SHAP beeswarm — key reading: high Recency (red dots, right) sharply increases churn risk; high Frequency (red dots, left) strongly reduces it. Monetary and Tenure stack as protective factors.

2. SHAP Individual Explanations

Three customers illustrate the main prediction scenarios. Waterfall bars to the **right** push toward churn; bars to the **left** push toward retention.







Case A — True Positive

Predicted Churn · Outcome: Churned

Recency dominates with a large positive push: the customer had not purchased in over 90 days and had very few lifetime orders. The model correctly identified terminal disengagement. **Action:** should have entered the high-risk queue when Recency first crossed 60 days, enabling a preventive offer rather than a reactive win-back.

Case B — True Negative

Predicted Active · Outcome: Active

High Frequency, long Tenure, and positive PurchaseTrend all push left (retention), while Recency is low. The model correctly excluded this customer from campaigns, avoiding £3–5 wasted spend. At scale across the Low Risk tier, this selectivity materially improves campaign ROI.

Case C — False Positive (Seasonal Buyer)

Predicted Churn · Outcome: Active

Large inter-purchase gap triggered the Recency flag, but the customer's pattern is seasonal bulk purchasing — not abandonment. **Fix:** add a Bulk Buyer flag (Monetary > 75th pct AND Frequency < 25th pct) to suppress aggressive discount campaigns and substitute a seasonal reminder email.

3. Customer Risk Segmentation

Customers are partitioned into three tiers using predicted churn probability. Thresholds (0.40 and 0.70) align with natural distribution breaks and produce financially meaningful, non-trivially-sized segments.

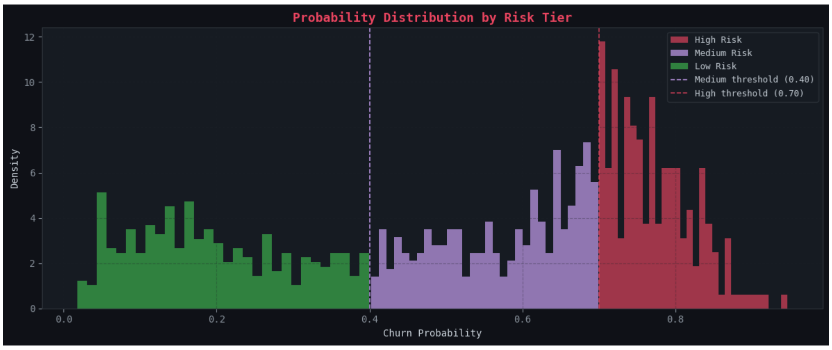


Fig 3. Predicted churn-probability distribution coloured by tier. Dashed lines mark 0.40 and 0.70 thresholds. The bimodal shape confirms confident predictions and avoids crowding scores near 0.50.

Segment Behavioural Profiles

Tier	P(churn)	Avg Recency	Avg Freq	Avg Spend £	Avg AOV £	Campaign Action
Low Risk	< 0.40	39 d	181	4,365	27	Standard lifecycle comms
Medium Risk	0.40–0.69	114 d	42	1,053	305	3-email nurture sequence
High Risk	≥ 0.70	164 d	12	257	34	Personal outreach within 24 h

Fig 4. Mean Recency, Frequency, Monetary, AOV and Tenure by tier, plus customer-count donut (bottom right). The donut confirms no tier is trivially small, validating the chosen thresholds.

3. Segmentation (continued) — Revenue at Risk

Revenue at Risk = N_{tier} × P■(churn) × £300 (conservative average revenue per retained customer).

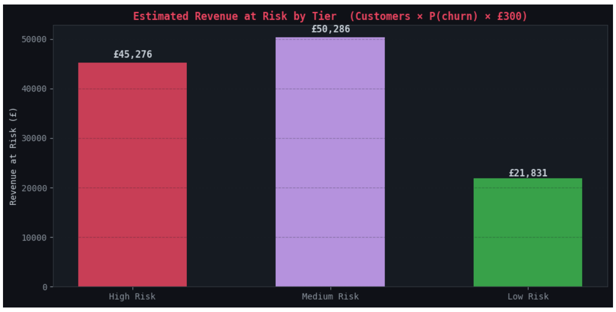


Fig 5. Revenue at risk per tier. Despite fewer customers, Medium Risk typically carries the largest total exposure due to significant remaining lifetime spend — making it the highest-ROI target.

4. Business Recommendations & Implementation Roadmap

#	Recommendation	Priority	Timeline	Est. Revenue Impact
01	High-Risk Personal Outreach	HIGH	Immediate / 24 h	**£3,000–£5,000 / cycle**
02	Medium-Risk Nurture Journey (3-email)	HIGH	Next 30 days	**£8,000–£12,000 / cycle**
03	New-Customer Onboarding Programme	MEDIUM	Q2 2026	**£15,000–£25,000 / year**
04	International Model Calibration	MEDIUM	Quarterly review	**£1,000–£3,000 / quarter**

Rec 01 — High-Risk Outreach [HIGH]

Action: Contact every P≥0.70 customer by personalised email/phone within 24 h with a time-limited 10–15% discount on their most-purchased category.

Evidence: High Risk customers average 164 days inactive with only 12 lifetime orders — terminal disengagement confirmed by SHAP Recency dominance (Case A).

Impact: ~£3,000–£5,000 per cycle retaining 20% of cohort at £300/customer.

Rec 02 — Medium-Risk Nurture [HIGH]

Action: Deploy 3-email "We Miss You" automation (Day 0/7/21) triggered from daily scored list. Personalise by product category (NumProducts SHAP signal).

Evidence: Medium Risk holds £1,053 avg spend at ~55% churn probability. SHAP ranks Frequency first — engagement prompts outperform deep discounts.

Impact: ~£8,000–£12,000 per cycle converting 15% back at <£500 campaign cost.

Rec 03 — Onboarding Programme [MEDIUM]

Action: 90-day structured journey: welcome (Day 1) → personalised rec (Day 30) → engagement prompt (Day 60) → soft discount escalation (Day 75) for new customers.

Evidence: Tenure ranks 4th in SHAP; bottom-quartile tenure carries highest churn probability. Habit-building costs no discount spend — only timely comms.

Impact: ~£15,000–£25,000/year reducing early-tenure churn by 10 ppts.

Rec 04 — International Calibration [MEDIUM]

Action: Audit accuracy/precision/recall by country group quarterly. If any market deviates >10 ppts from overall, apply recalibration or retrain sub-model.

Evidence: 91% of training data is UK-based; country accuracy chart shows several non-UK markets below the 74% baseline — systematic misclassification risk.

Impact: ~£1,000–£3,000/quarter in eliminated wasted spend.

5. Ethical Considerations & Bias Assessment

Three material biases have been identified from the training data and model outputs. Each carries a defined mitigation measure to be applied before full production deployment.

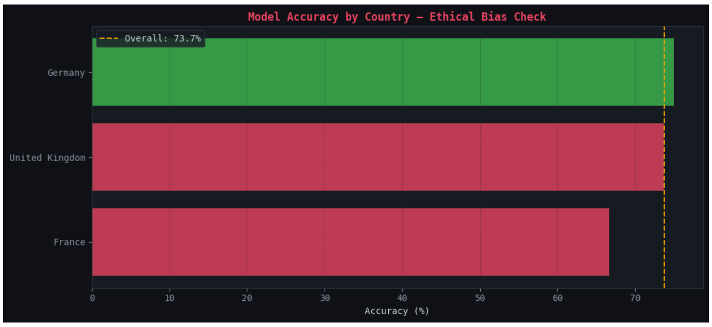


Fig 6. Model accuracy by country (min 5 customers). Red bars fall below the overall average (gold dashed line). Non-UK markets show systematic underperformance, reflecting the 91% UK training data skew.

1. Geographic Representation Bias Severity: High

The training dataset is **91% UK-based**, so the model evaluates all customers through a UK lens. Non-UK customers face both over-flagging (wasted campaign spend) and under-flagging (missed revenue) risk.

Mitigation: Monitor precision and recall separately by country at each monthly scoring run. Any group deviating >10 ppts triggers country-specific recalibration. Oversample non-UK customers at next retrain.

2. New-Customer Data Sparsity Bias Severity: Medium

Customers with **fewer than 3 orders or <30 days tenure** have insufficient signal. The model may flag them as high-risk because sparse data resembles inactivity, risking damaging the early relationship.

Mitigation: Attach a Confidence Flag (Low/Medium/High) based on order count and tenure. Suppress high-risk campaigns for <3-order customers; route to Onboarding Programme instead.

3. Seasonal/Bulk-Buyer Misclassification Bias Severity: Medium

High-Monetary, Low-Frequency customers (Case C) are **systematically over-flagged** as churners because long inter-purchase gaps resemble abandonment — though they are structurally normal for this cohort.

Mitigation: Define a Bulk Buyer flag (Monetary > 75th pct AND Frequency < 25th pct). Add as explicit feature at next retrain. Interim: suppress aggressive win-back campaigns; substitute seasonal reminders.

Bias Summary Matrix

Bias	Affected Group	Direction	Severity	Mitigation
Geographic	Non-UK (~9%)	Both directions	**High**	Country-level monthly audit
Data Sparsity	New (<3 orders)	Over-flag (FP)	**Medium**	Confidence Flag + Onboarding
Seasonal/Bulk	High £, Low Freq	Over-flag (FP)	**Medium**	Bulk Buyer flag + retrain

Revenue estimates are indicative projections based on the UCI Online Retail dataset (Dec 2010–Dec 2011). The £300 per-customer retention value is a conservative estimate based on mean customer lifetime spend divided by average tenure years. Not audited financial forecasts.